

COMBINATORICS

CATALAN NUMBERS HOMEWORK

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Theorem 1. *Let C_n represent the Catalan numbers, with*

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

. We define the Catalan numbers as the number of valid ways to arrange n -pairs of left-right parentheses where every opening bracket has a subsequent closing bracket, and they are nested correctly. We want to show that C_n as defined above, also represents the number of binary trees with n vertices.

Proof. Our proof technique is the mechanics of reduction. We reduce one problem to another based on definition. Our goal will be to show that there is a bijective correspondence between the two definitions and their outcomes.

The fundamental idea behind the proof will be that every binary tree is determined by a root and a pair of smaller trees. So counting binary trees in this fashion is a split of counting two independent sub-Catalan objects.

Suppose, Φ defines a map from binary trees to balanced parentheses strings. We do the following iteratively: (a) the empty tree maps to the empty string, and (b) if a tree has a root, a left subtree L , and a right subtree R , then we write $\Phi(T) = (\Phi(L))\Phi(R)$. So each vertex contributes one pair of matched parentheses with the left subtree written inside the pair and the right subtree afterward.

Why is this a bijection?

Every binary tree has a decomposition into: (a) a root, (b) a left subtree, and (c) a right subtree. Every non-empty balanced parentheses string has a unique decomposition of the form

$$(w_0)w_1,$$

where w_0 and w_1 are balanced parentheses strings.

So, from a balanced string w we recover a tree by using the following steps: (a) making a root, (b) taking the substring inside the first matching pair as the left subtree, and (c) taking the remaining suffix substring as the right subtree. Thus, we see that Φ is one-to-one and onto, a bijection.

Since the balanced parentheses strings with n -pairs are counted by the Catalan numbers C_n , then it naturally follows that the binary trees on n -vertices are counted by C_n . $\square$